

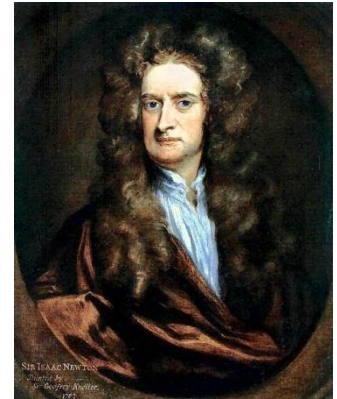
Quantum Chemistry Basics

Dr. Mahmoud A. Noamaan

**School of computational quantum chemistry
Cairo University, Faculty of science,
mathematics Department**

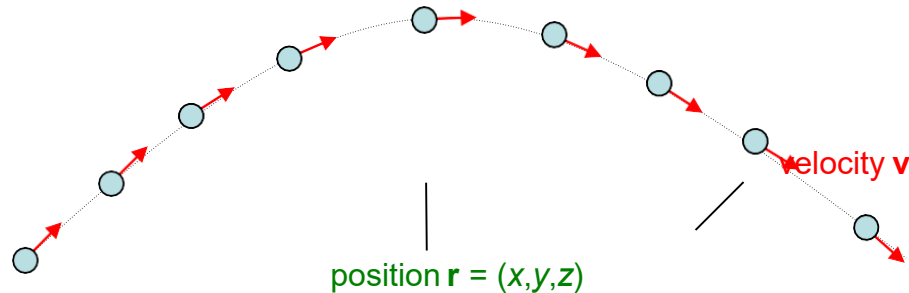
1. Classical Mechanics

- *Do the electrons in atoms and molecules obey Newton's classical laws of motion?*
- We shall see that the answer to this question is “**No**”.
- This has led to the development of **Quantum Mechanics** – we will contrast classical and quantum mechanics.



1.1 Features of Classical Mechanics (CM)

1) CM predicts a precise trajectory for a particle.



- The exact position ($\mathbf{r}$) and velocity ($\mathbf{v}$) (and hence the momentum $\mathbf{p} = m\mathbf{v}$) of a particle (mass = m) can be known simultaneously at each point in time.
- **Note:** position ($\mathbf{r}$), velocity ($\mathbf{v}$) and momentum ($\mathbf{p}$) are vectors, having magnitude **and** direction $\Rightarrow \mathbf{v} = (v_x, v_y, v_z)$.
- objects are treated as particles with well-defined positions and momenta
- behaviour is governed by Newton's equations (or equivalent):
$$\mathbf{F} = m\mathbf{a}$$
- appropriate for objects ranging from atoms to planets

- 2) Any type of motion (translation, vibration, rotation) can have any value of energy associated with it
– i.e. there is a continuum of energy states.
- 3) Particles and waves are distinguishable phenomena, with different, characteristic properties and behaviour.

	Property		Behaviour
Particles	$\left\{ \begin{array}{l} \text{mass} \\ \text{position} \\ \text{velocity} \end{array} \right\}$	$\Rightarrow$	$\left\{ \begin{array}{l} \text{momentum} \\ \text{collisions} \end{array} \right\}$
Waves	$\left\{ \begin{array}{l} \text{wavelength} \\ \text{frequency} \end{array} \right\}$	$\Rightarrow$	$\left\{ \begin{array}{l} \text{diffraction} \\ \text{interference} \end{array} \right\}$

1.2 Revision of Some Relevant Equations in CM

Total energy of particle:

$$E = \text{Kinetic Energy (KE)} + \text{Potential Energy (PE)}$$

T - depends on **v**

V - depends on **r**

V depends on the system
e.g. positional, electrostatic PE

$$E = \frac{1}{2}mv^2 + V$$

$$\Rightarrow E = \frac{p^2}{2m} + V \quad (p = mv)$$

Note: strictly E, T, V (and **r**, **v**, **p**) are all defined at a particular time (**t**) – E(t) etc..

- Consider a 1-dimensional system (straight line translational motion of a particle under the influence of a potential acting parallel to the direction of motion):

- Define:

position	$r = x$
velocity	$v = dx/dt$
momentum	$p = mv = m(dx/dt)$
PE	V
force	$F = -(dV/dx)$

- Newton's 2nd Law of Motion

$$F = m \underset{\substack{| \\ \text{acceleration}}}{a} = m(dv/dt) = m(d^2x/dt^2)$$

- Therefore, if we know the forces acting on a particle we can solve a differential equation to determine its trajectory $\{x(t), p(t)\}$.

2. Quantum Mechanics Review

Quantum Mechanics (Heisenberg, Schrödinger, *et al.*)

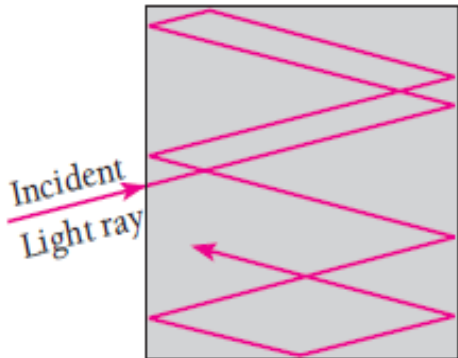
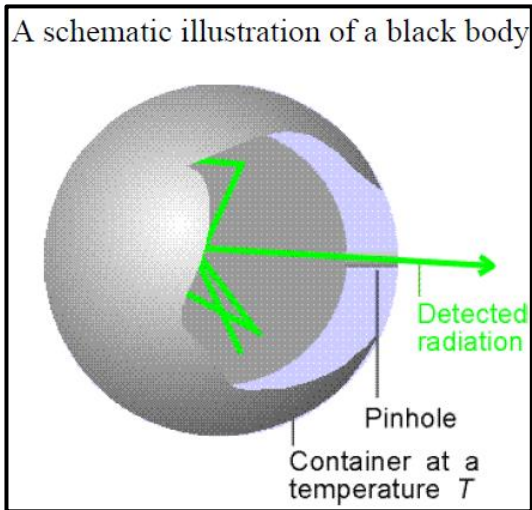
- objects are described as probability distributions
- cannot know exactly a particle's position *and* momentum simultaneously
→ Heisenberg uncertainty principle
- all information regarding the system is contained in the wavefunction
→ wavefunction obtained through Schrödinger equation
(or equivalent)
- needed to describe behaviour of small, light particles (electrons, maybe protons)

Failures of Classical Physics

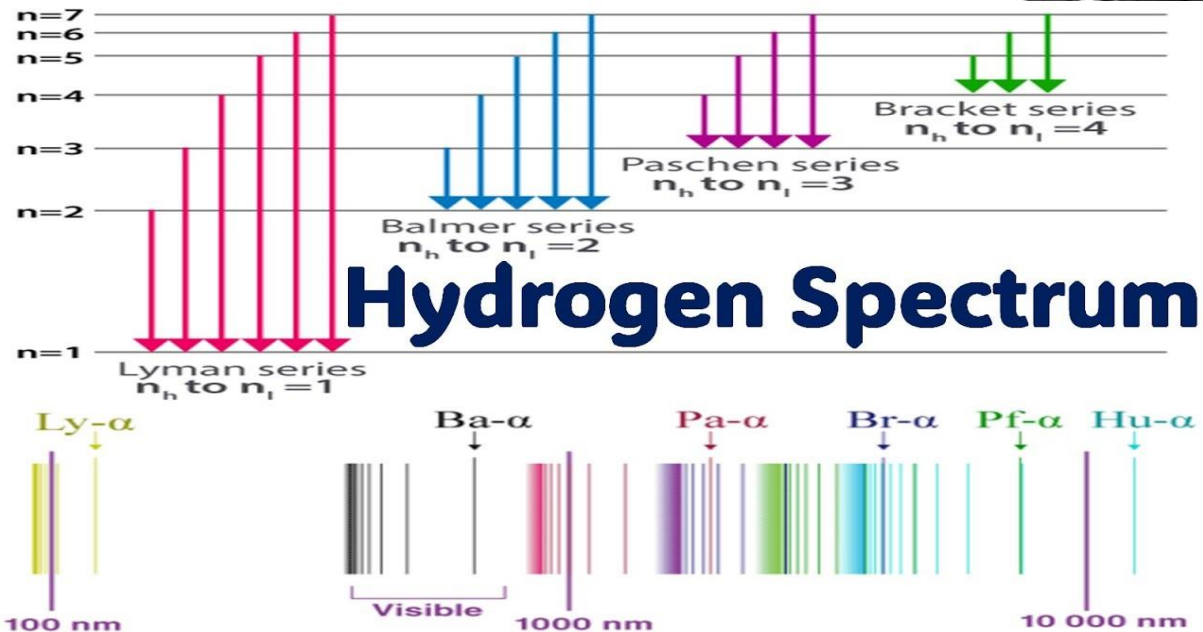
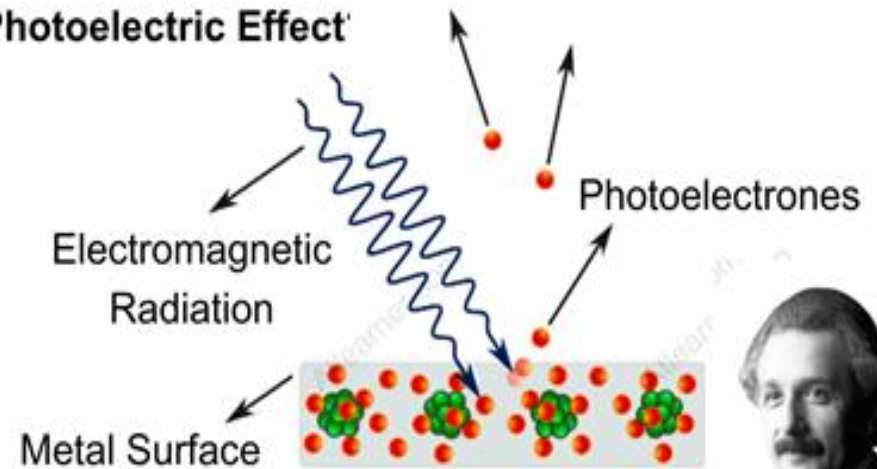
The Blackbody Radiation

■ when you heat a body or metal it emits light

A schematic illustration of a black body



Photoelectric Effect



Bohr Model and Limitations

Bohr's model introduced quantized electron orbits to explain hydrogen's spectrum.

However, it couldn't describe multi-electron atoms or molecular bonding accurately.

Quantum Chemical Methods

quantum chemical methods employ a quantum mechanical treatment of the electrons without resorting to parameterization

also called:

- *ab initio*
- first principles
- electronic structure methods
- molecular orbital calculations

What do we get from these calculations?

- electronic wavefunction → contains all information regarding the system
- **can calculate electronic energy**
 - **useful for exploring the potential energy surface**
 - **needed to calculate reaction energies and barriers**
- **can describe changes in electronic structure/bonds**
 - **can study changes in molecular orbitals**
- **can calculate other properties related to the electronic structure**
 - **various spectra**
 - **can approximate quantities like atomic charges and bond orders**
 - **reactivity indices**

Postulates of Quantum Mechanics

Postulate 1:

- the state of a system is described by a function $\Psi(\mathbf{r}, t)$ called a wavefunction
- $\Psi(\mathbf{r}, t)$ is single-valued, continuous, quadratically integrable
- the probability that the particle is in volume $d\tau$ at position $\mathbf{r}$ and time t is:

$$\Psi^* (\mathbf{r}, t) \Psi (\mathbf{r}, t) d\tau$$

- since the particle must be somewhere all the time:

$$\int \Psi^* (\mathbf{r}, t) \Psi (\mathbf{r}, t) d\tau = 1$$

* indicates the complex conjugate of Ψ

- $i \rightarrow -i$, e.g.:

$$\Psi = e^{-i2x} \rightarrow \Psi^* = e^{i2x}$$

$$i = \sqrt{-1}$$

$d\tau$ indicates integration over all coordinates

bold fonts indicate vectors
e.g. $\mathbf{r} = (x, y, z)$

Postulates of Quantum Mechanics

Postulate 1:

- the state of a system is described by a function $\Psi(\mathbf{r}, t)$ called a wavefunction
- $\Psi(\mathbf{r}, t)$ is single-valued, continuous, quadratically integrable
- the probability that the particle is in volume $d\tau$ at position $\mathbf{r}$ and time t is:

$$\Psi^*(\mathbf{r}, t) \Psi(\mathbf{r}, t) d\tau$$

- since the particle must be somewhere all the time:

$$\int \Psi^*(\mathbf{r}, t) \Psi(\mathbf{r}, t) d\tau = 1$$

Postulate 2:

- every physical observable is represented by a linear Hermitian operator

Linear Hermitian Operators

- Operator:
- a rule that turns a function into another function
 - designated with a „hat“ (sometimes I’ll use *italics*)

$$\hat{D} = \frac{\partial}{\partial x}; f(x) = x^2 \Rightarrow \hat{D}f(x) = \frac{\partial x^2}{\partial x} = 2x$$

- Linear Operator:
- an operator with the following properties

$$\hat{O}[f(x) + g(x)] = \hat{O}f(x) + \hat{O}g(x)$$

$$\hat{O}[cf(x)] = c\hat{O}f(x)$$

- Hermitian Operator:
- an operator with the following property

$$\int f^* \hat{O}g d\tau = \int g (\hat{O}f)^* d\tau$$

- guarantees real values for associated property

Postulates of Quantum Mechanics

- Postulate 2:
- every physical observable is represented by a linear Hermitian operator
 - quantum mechanical operators are analogues of classical expressions using:

$$x \rightarrow x \qquad p_x \rightarrow -i\hbar \frac{\partial}{\partial x}$$

- for example, consider the kinetic energy:

- classically:
$$T = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2)$$

- quantum mechanically:
$$\hat{T} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) = -\frac{\hbar^2}{2m} \nabla^2$$

Postulates of Quantum Mechanics

- Postulate 3:
- in any measurement of the observable associated with operator B the only values that will ever be observed are eigenvalues b_i that satisfy:

$$\hat{B}\Psi_i = b_i\Psi_i$$

eigenvalue of B

eigenfunction of B

- also called eigenstate

- Ψ_i is an eigenfunction of operator B if operating on Ψ_i with B gives back Ψ_i multiplied by a constant
- constant is called an eigenvalue, value of property B when system is in state i

example:

$$\hat{B} = \frac{\partial}{\partial x}$$

$$\Psi = e^{-2x}$$

$$\hat{B}\Psi = \frac{\partial e^{-2x}}{\partial x} = -2e^{-2x}$$

Eigenvalues and Eigenfunctions

- eigenfunctions of Hermitian operators are orthonormal

$$\int \Psi_i^* \Psi_j d\tau = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

- basically says system is in state i or j , but not both

- lets us calculate the eigenvalue for a given state

$$\hat{B}\Psi_i = b_i\Psi_i \Rightarrow \int \Psi_j^* \hat{B}\Psi_i d\tau = \int \Psi_j^* b_i \Psi_i d\tau = b_i \int \Psi_j^* \Psi_i d\tau = b_i \delta_{ij}$$

$$b_i = \int \Psi_i^* \hat{B}\Psi_i d\tau$$

Postulates of Quantum Mechanics

- Postulate 3:
- in any measurement of the observable associated with operator B the only values that will ever be observed are eigenvalues b_i that satisfy:

$$\hat{B}\Psi_i = b_i\Psi_i$$

- this postulate introduces notion of discrete states, which is at the heart of quantum mechanics

- Postulate 4:
- an arbitrary state Ψ can be represented as a linear combination of the eigenfunctions of a quantum mechanical operator

$$\Psi = \sum_i c_i \Psi_i$$

- called superposition principle
- system must adopt a specific state upon measurement

Postulates of Quantum Mechanics

- Postulate 5: • the average value of property B at time t is:

$$\langle B \rangle = \int \Psi^* \hat{B} \Psi d\tau$$

- Postulate 6: • time development of a state is given by the time-dependent Schrödinger equation:

$$-\frac{\hbar}{i} \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = \hat{H} \Psi(\mathbf{r}, t)$$

- H is the Hamiltonian = total energy operator

$$E = T + V \Rightarrow \hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \hat{V}(\mathbf{r}, t)$$

Time-Independent Schrödinger Equation

for most chemical problems V is independent of time

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \hat{V}(\mathbf{r}) \longleftarrow \text{usually Coulomb potential}$$

$$\hat{V} = \frac{q_1 q_2}{4\pi\epsilon_0 r_{12}}$$

- energy of state specified by Ψ is constant at all times

$$\hat{H}\Psi(\mathbf{r}, t) = E\Psi(\mathbf{r}, t)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(\mathbf{r}, t)}{\partial t^2} = \hat{H}\Psi(\mathbf{r}, t) \Rightarrow \Psi(\mathbf{r}, t) = e^{-iEt/\hbar} \psi(\mathbf{r})$$

- $\psi(\mathbf{r})$ is the solution to the time-independent Schrödinger equation

$$\hat{H}\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

time-independent Hamiltonian $\nearrow$

$\nwarrow$ time-independent wavefunction

Particles in a 1D or 2D box

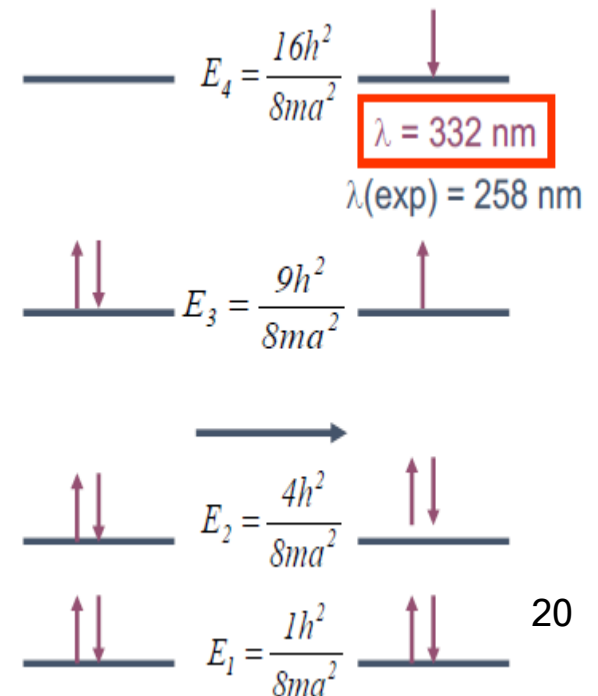
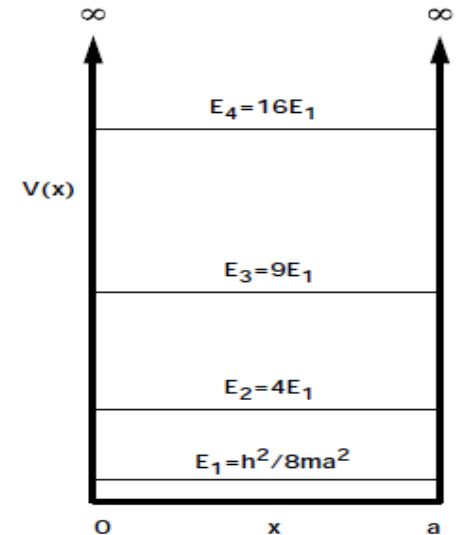
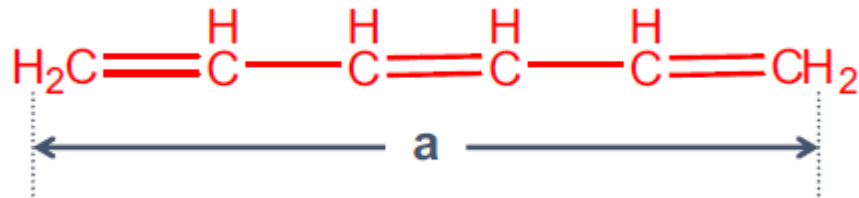
$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x)$$

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & x < 0 \text{ and } x > a \end{cases}$$

$$E_n = \frac{\hbar^2 \pi^2}{2ma^2} n^2 = \frac{h^2}{8ma^2} n^2 \quad n = 1, 2, 3 \dots$$

$$\psi_n(x) = A \sin \frac{n\pi x}{a}, \quad n = 1, 2, 3 \dots$$

Free Electron Molecular Orbital (FEMO)



Hydrogen Atom

$$\hat{H} \psi = E \psi$$

$$-(\hbar^2/2m \nabla^2 + Ze^2/4\pi\epsilon_0 r) \psi = E \psi$$

r, θ and ϕ

the wavefunction takes the form

$$\Psi(r, \theta, \phi) = R(r)P(\theta)F(\phi)$$

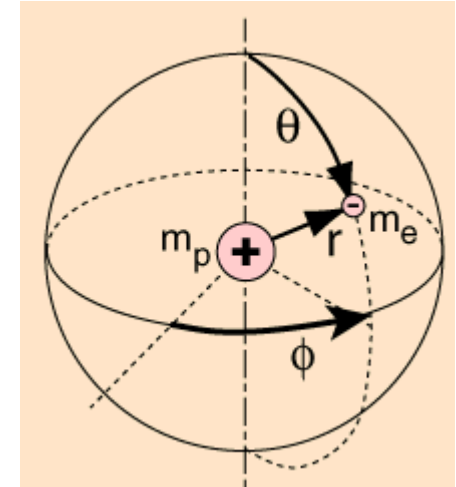
$R(r)$ [Principal quantum number](#) $n = 1, 2, 3, \dots$

$P(\theta)$ [Orbital quantum number](#) $\ell = 0, 1, 2, \dots, n-1$

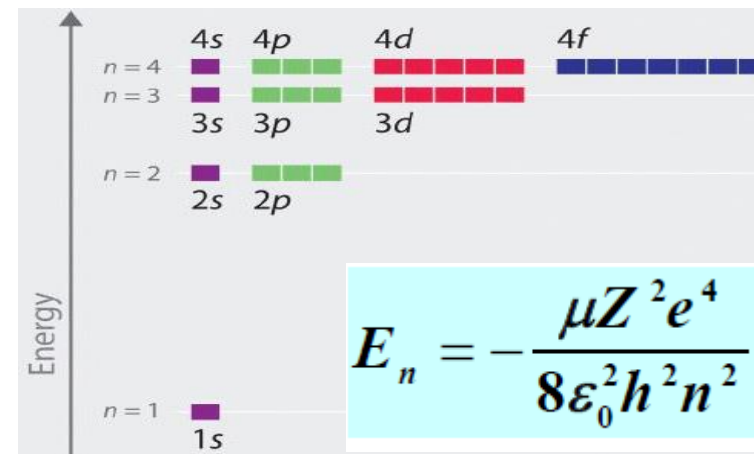
$F(\phi)$ [Magnetic quantum number](#)

$m_\ell = -\ell, -\ell+1, \dots, 0, \dots, \ell-1, \ell$ or $2\ell+1$ values

$\uparrow \downarrow$ [Spin quantum number](#) $m_s = +\frac{1}{2}, -\frac{1}{2}$



Laplacian: $\nabla^2 F = \Delta^2 F = \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} + \frac{\partial^2 F}{\partial z^2}$



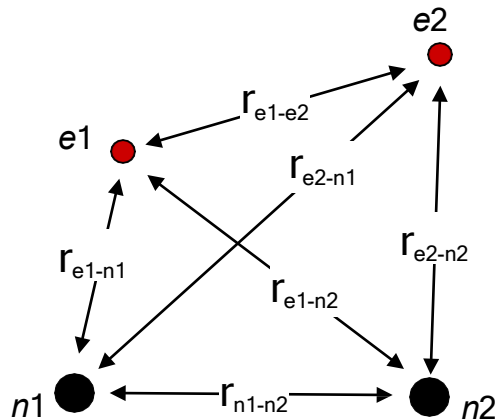
Molecular Hamiltonian

quantum chemical methods strive to solve the time-independent Schrödinger equation

$$\hat{H}\Psi(\mathbf{r}) = E\Psi(\mathbf{r})$$

for a system of N electrons in the presence of M nuclei using only mathematical approximations

→ no parameterization/fitting to experimental data



2 nuclei, 2 electrons

terms included in the Hamiltonian

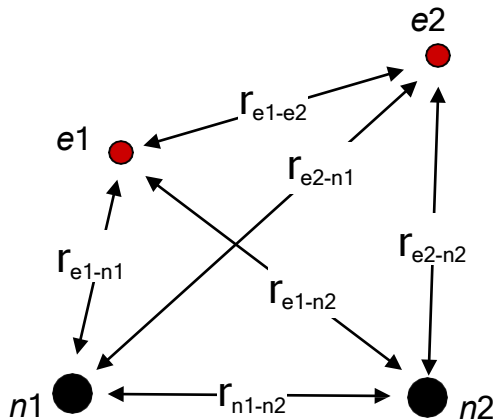
1 electron-electron interaction

1 nucleus-nucleus interaction

4 nucleus-electron interactions

kinetic energy of electrons and nuclei

Molecular Hamiltonian



2 nuclei, 2 electrons

terms included in the Hamiltonian

1 electron-electron interaction

1 nucleus-nucleus interaction

4 nucleus-electron interactions

kinetic energy of electrons and nuclei

$$\hat{H} = \underbrace{-\sum_{i=1}^N \frac{\hbar^2}{2m_e} \nabla_i^2}_{\text{electron kinetic energy}} - \underbrace{\sum_{I=1}^M \frac{\hbar^2}{2M_I} \nabla_I^2}_{\text{nuclear kinetic energy}} - \underbrace{\sum_{I=1}^M \sum_{i=1}^N \frac{Z_I e^2}{4\pi\epsilon_0 r_{Ii}}}_{\text{nuclear-electron attraction}} + \underbrace{\sum_{i=1}^{N-1} \sum_{j>i}^N \frac{e^2}{4\pi\epsilon_0 r_{ij}}}_{\text{electron-electron repulsion}} + \underbrace{\sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J e^2}{4\pi\epsilon_0 r_{IJ}}}_{\text{nuclear-nuclear repulsion}}$$

simplifications:

Laplacian: $\nabla^2 \mathbf{F} = \Delta^2 \mathbf{F} = \frac{\partial^2 \mathbf{F}}{\partial x^2} + \frac{\partial^2 \mathbf{F}}{\partial y^2} + \frac{\partial^2 \mathbf{F}}{\partial z^2}$

1. Born-Oppenheimer approximation
2. atomic units

Born-Oppenheimer Approximation

$$\hat{H} = \underbrace{-\sum_{i=1}^N \frac{\hbar^2}{2m_e} \nabla_i^2}_{\text{electron kinetic energy}} - \underbrace{\sum_{I=1}^M \frac{\hbar^2}{2M_I} \nabla_I^2}_{\text{nuclear kinetic energy}} - \underbrace{\sum_{I=1}^M \sum_{i=1}^N \frac{Z_I e^2}{4\pi\epsilon_0 r_{Ii}}}_{\text{nuclear-electron attraction}} + \underbrace{\sum_{i=1}^{N-1} \sum_{j>i}^N \frac{e^2}{4\pi\epsilon_0 r_{ij}}}_{\text{electron-electron repulsion}} + \underbrace{\sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J e^2}{4\pi\epsilon_0 r_{IJ}}}_{\text{nuclear-nuclear repulsion}}$$

- nuclei are much heavier than electrons

- $m_{\text{proton}} \sim 2000 m_{\text{electron}}$
- nuclei move much more slowly
- consider electrons as moving in field of fixed nuclei

- implications

- nuclear kinetic energy can be neglected
- nuclear-nuclear repulsion energy is constant

$$\hat{H} = \underbrace{-\sum_{i=1}^N \frac{\hbar^2}{2m_e} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I e^2}{4\pi\epsilon_0 r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{e^2}{4\pi\epsilon_0 r_{ij}}}_{\text{terms contributing to electronic energy} = \hat{H}_{\text{elec}}} + \underbrace{\sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J e^2}{4\pi\epsilon_0 r_{IJ}}}_{\text{nuclear-nuclear repulsion added as a constant}}$$

Born-Oppenheimer Approximation

$$\hat{H} = - \underbrace{\sum_{i=1}^N \frac{\hbar^2}{2m_e} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I e^2}{4\pi\epsilon_0 r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{e^2}{4\pi\epsilon_0 r_{ij}}}_{\text{terms contributing to electronic energy} = \hat{H}_{elec}} + \underbrace{\sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J e^2}{4\pi\epsilon_0 r_{IJ}}}_{\text{nuclear-nuclear repulsion added as a constant}}$$

- solution to Schrodinger equation for electronic Hamiltonian:

$$\hat{H}_{elec} \Psi = E_{elec} \Psi$$

- is the electronic wavefunction:

$$\Psi = \Psi_{elec} (\{\mathbf{r}_i\}; \{\mathbf{R}_I\})$$

- depends explicitly on electronic coordinates, $\mathbf{r}_i$, and parametrically on the nuclear coordinates, $\mathbf{R}_I$
- total energy also depends parametrically on $\mathbf{R}_I$:

$$E_{tot} = E_{elec} (\{\mathbf{R}_I\}) + \sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J e^2}{4\pi\epsilon_0 r_{IJ}}$$

- relationship between E_{tot} and $\{\mathbf{R}_I\}$ yields the potential energy surface

Atomic Units

$$\hat{H} = - \underbrace{\sum_{i=1}^N \frac{\hbar^2}{2m_e} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I e^2}{4\pi\epsilon_0 r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{e^2}{4\pi\epsilon_0 r_{ij}}}_{\text{terms contributing to electronic energy} = \hat{H}_{elec}} + \underbrace{\sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J e^2}{4\pi\epsilon_0 r_{IJ}}}_{\text{nuclear-nuclear repulsion added as a constant}}$$

simplify by changing units:

$$\hbar = m_e = e = 4\pi\epsilon_0 = 1$$

$$\text{distance: } a_0 = \text{bohr radius} = 5.2918 \times 10^{-11} \text{ m}$$

$$\text{energy: Hartree} = 4.3598 \times 10^{-18} \text{ J} = 627.51 \text{ kcal/mol}$$

$$\hat{H} = - \underbrace{\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{1}{r_{ij}}}_{\text{terms contributing to electronic energy} = \hat{H}_{elec}} + \underbrace{\sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J}{r_{IJ}}}_{\text{nuclear-nuclear repulsion added as a constant}}$$

Quantum Chemical Methods

we want to calculate the total molecular energy using:

$$\hat{H}_{tot} \Psi = E_{tot} \Psi$$

where H_{tot} is:

$$\hat{H}_{tot} = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{1}{r_{ij}} + \sum_{I=1}^{M-1} \sum_{J>I}^M \frac{Z_I Z_J}{r_{IJ}}$$

since the last term is a constant, we really need to solve:

$$\hat{H}_{elec} \Psi = E_{elec} \Psi$$

where H_{elec} is:

$$\hat{H}_{elec} = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{1}{r_{ij}}$$

- will give us the electronic energy
- will give us the electronic wavefunction → used to get other properties

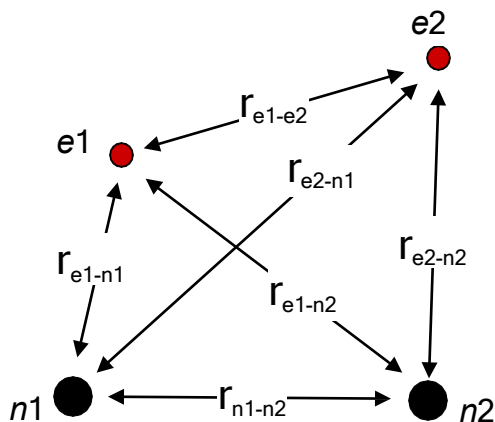
Quantum Chemical Methods

it is not possible to solve the Schrodinger equation exactly for a many-body system

$$\hat{H}_{elec} = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{1}{r_{ij}}$$

electron-electron terms are the problem

called the many-body problem



- the motion of electron 1 affects the motion of electron 2, and vice-versa
- prevents separability of the Hamiltonian
 - means we can't break down the Hamiltonian into independent equations for electrons 1 and 2
- many-body problem is not unique to quantum mechanics and electrons

Quantum Chemical Methods

it is not possible to solve the Schrodinger equation exactly for a many-body system

$$\hat{H}_{elec} = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{1}{r_{ij}}$$

electron-electron terms
are the problem

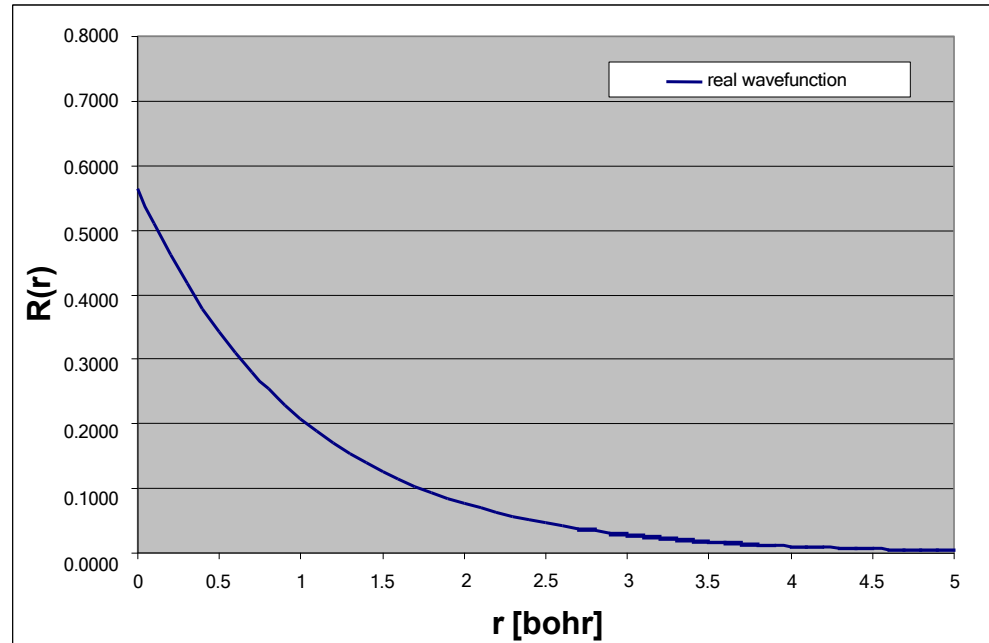
Strategies used in quantum chemistry:

1. treat electron-electron interactions as a minor perturbation
 - called **perturbation theory**
 - more on this later
2. „guess“ a suitable form of Ψ and optimize it
 - called **variational method**
 - basis for most of quantum chemistry
 - we'll look at this now

Variational Method

Example for the Hydrogen atom:

$$\text{real 1s: } \Psi = \left(\frac{1}{\pi} \right)^{1/2} e^{-r}$$



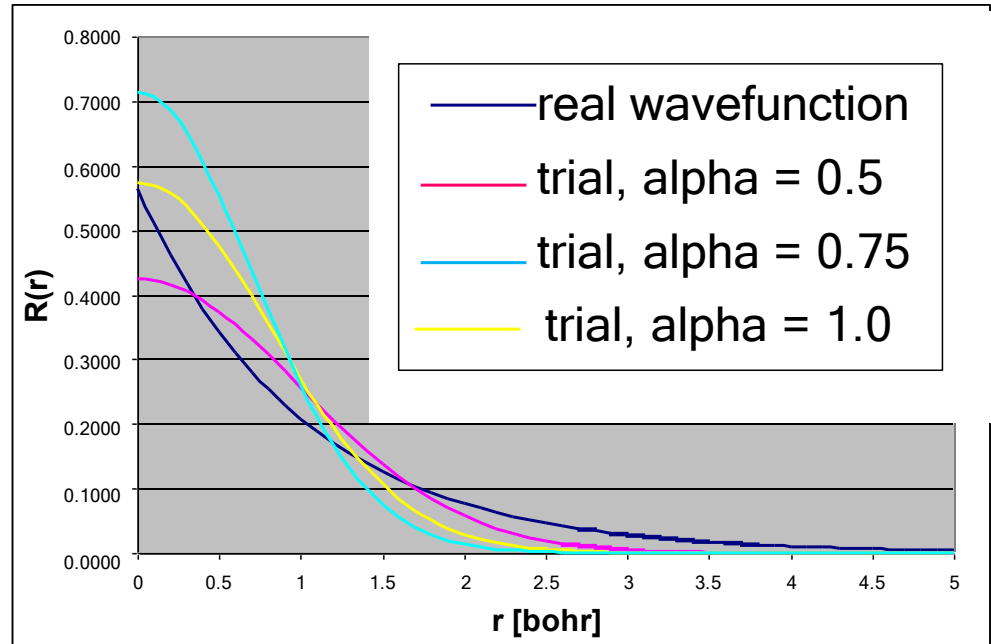
Variational Method

Example for the Hydrogen atom:

$$\text{real 1s: } \Psi = \left(\frac{1}{\pi} \right)^{1/2} e^{-r}$$

$$\text{trial 1s: } \Phi = \left(\frac{2\alpha}{\pi} \right)^{3/4} e^{-\alpha r^2}$$

we want to pick α to give the best agreement with the real solution



Obvious questions:

1. can we really do this?
2. how do we „optimize“ Φ (i.e. how do we pick α)?
3. what form of Φ should we use?

Variational Principle - Some Background

there exists an infinite set of wavefunctions, $\{\Psi_\alpha\}$, such that:

$$\hat{H}\Psi_\alpha = E_\alpha \Psi_\alpha \quad \alpha = 0, 1, 2, \dots$$

where the energies E_α are the eigenvalues for a particular state:

$$E_0 \leq E_1 \leq E_2 \dots \leq E_\alpha \dots$$

since the wavefunctions are orthonormal: $\int \Psi_i^* \Psi_j d\tau = \delta_{ij}$

$$\int \Psi_i^* \hat{H} \Psi_j d\tau = \int \Psi_i^* E_j \Psi_j d\tau$$

$$\int \Psi_i^* \hat{H} \Psi_j d\tau = E_j \delta_{ij} \quad \text{for } i = j \quad \delta_{ij} = 1$$

$$E_j = \int \Psi_j^* \hat{H} \Psi_j d\tau$$

Variational Principle - Some Background

$$E_j = \int \Psi_j^* \hat{H} \Psi_j d\tau \longrightarrow \text{energy of a specific state}$$

- we can't solve this because we don't know what Ψ_j is

variational approach (basic idea):

- let's take a guess at the form of Ψ_j and try to get as close as possible to the real wavefunction
- trial wavefunction must meet certain constraints to be valid
 - more on this later

4th postulate of quantum mechanics:

- an arbitrary wavefunction can be expressed as a linear combination of the set $\{\Psi_i\}$

$$\Phi = \sum_i c_i \Psi_i$$

Variational Principle - Some Background

trial wavefunction: $\Phi = \sum_i c_i \Psi_i$

- choose Φ to be orthonormal
- don't know exact form of $\{\Psi_i\}$

$$\begin{aligned}\int |\Phi|^2 d\tau &= 1 = \int \sum_i c_i^* \Psi_i^* \sum_j c_j \Psi_j d\tau \\&= \sum_i \sum_j c_i^* c_j \int \Psi_i^* \Psi_j d\tau \\&= \sum_i \sum_j c_i^* c_j \delta_{ij} \\&= \sum_i |c_i|^2 \quad \text{for } i = j \quad \delta_{ij} = 1 \\&\therefore \sum_i |c_i|^2 = 1\end{aligned}$$

Variational Principle - Some Background

trial wavefunction:

Let's consider evaluating the energy associated with the trial wavefunction Φ :

$$\Phi = \sum_i c_i \Psi_i \quad \int |\Phi|^2 d\tau = \sum_i |c_i|^2 = 1$$

- choose Φ to be orthonormal
- don't know exact form of $\{\Psi_i\}$

$$\int \Phi^* \hat{H} \Phi d\tau = \int \left(\sum_i c_i^* \Psi_i^* \right) \hat{H} \left(\sum_j c_j \Psi_j \right) d\tau$$

$$= \sum_i c_i^* \sum_j c_j \int \Psi_i^* \hat{H} \Psi_j d\tau$$

$$= \sum_i c_i^* \sum_j c_j E_j \delta_{ij}$$

$$= \sum_i |c_i|^2 E_i \quad \text{for } i = j \quad \delta_{ij} = 1$$

Thus, the energy associated with Φ is determinable from all the coefficients c_i and their associated energies E_i

Variational Principle - Some Background

what we know:

$$\Phi = \sum_i c_i \Psi_i \quad \int |\Phi|^2 d\tau = \sum_i |c_i|^2 = 1 \quad \int \Phi \hat{H} \Phi d\tau = \sum_i |c_i|^2 E_i$$

compare the trial energy with the ground state energy:

$$\int \Phi^* \hat{H} \Phi d\tau - E_0 \int |\Phi|^2 d\tau = \sum_i |c_i|^2 E_i - \sum_i |c_i|^2 E_0$$

Variational Principle - Some Background

what we know:

$$\Phi = \sum_i c_i \Psi_i \quad \int |\Phi|^2 d\tau = \sum_i |c_i|^2 = 1 \quad \int \Phi \hat{H} \Phi d\tau = \sum_i |c_i|^2 E_i$$

compare the trial energy with the ground state:

$$\int \Phi^* \hat{H} \Phi d\tau - E_0 \int |\Phi|^2 d\tau = \sum_i |c_i|^2 (E_i - E_0)$$

Variational Principle - Some Background

what we know:

$$\Phi = \sum_i c_i \Psi_i \quad \int |\Phi|^2 d\tau = \sum_i |c_i|^2 = 1 \quad \int \Phi \hat{H} \Phi d\tau = \sum_i |c_i|^2 E_i$$

compare the trial energy with the ground state:

$$\int \Phi^* \hat{H} \Phi d\tau - E_0 \int |\Phi|^2 d\tau = \sum_i |c_i|^2 (E_i - E_0)$$

Both $|c_i|^2$ and $E_i - E_0$ must be equal to or greater than zero:

$$\sum_i |c_i|^2 (E_i - E_0) \geq 0$$

Thus, we have
$$\int \Phi^* \hat{H} \Phi d\tau - E_0 \int |\Phi|^2 d\tau \geq 0$$

and with some rearrangement:

$$\frac{\int \Phi^* \hat{H} \Phi d\tau}{\int |\Phi|^2 d\tau} \geq E_0$$

- this is called the variational integral

Variational Principle

energy of trial function, Φ $\longrightarrow$ $\frac{\int \Phi^* \hat{H} \Phi d\tau}{\int |\Phi|^2 d\tau} \geq E_0$ $\longleftarrow$ true ground state energy of the system

normalization factor $\nearrow$ $\int |\Phi|^2 d\tau$

Variational Theorem

The expectation value of the energy is always greater than or equal to the true energy

for any suitable trial wavefunction:

- the energy of the trial function is guaranteed to be higher than the real ground state energy
- if we happen upon the real wavefunction we get E_0
- otherwise, we get an upper bound on E_0
- basic principle: trial wavefunction with lower energy is a „better“ trial wavefunction

Construction and Optimization of Trial Wavefunctions

$$\frac{\int \Phi^* \hat{H} \Phi d\tau}{\int |\Phi|^2 d\tau} \geq E_0 \quad \text{nice concept, but how do we use it?}$$

Linear variation:

1. pick a trial function that is a linear combination of basis functions

$$\Phi = \sum_{i=1}^N a_i \phi_i$$

← well-defined functions

← coefficients in linear expansion

2. modify coefficients $\{a_i\}$ to minimize the energy

the „best“ trial wavefunction gives the lowest energy

Construction and Optimization of Trial Wavefunctions

trial function is a linear combination of basis functions

$$\Phi = \sum_{i=1}^N a_i \phi_i$$

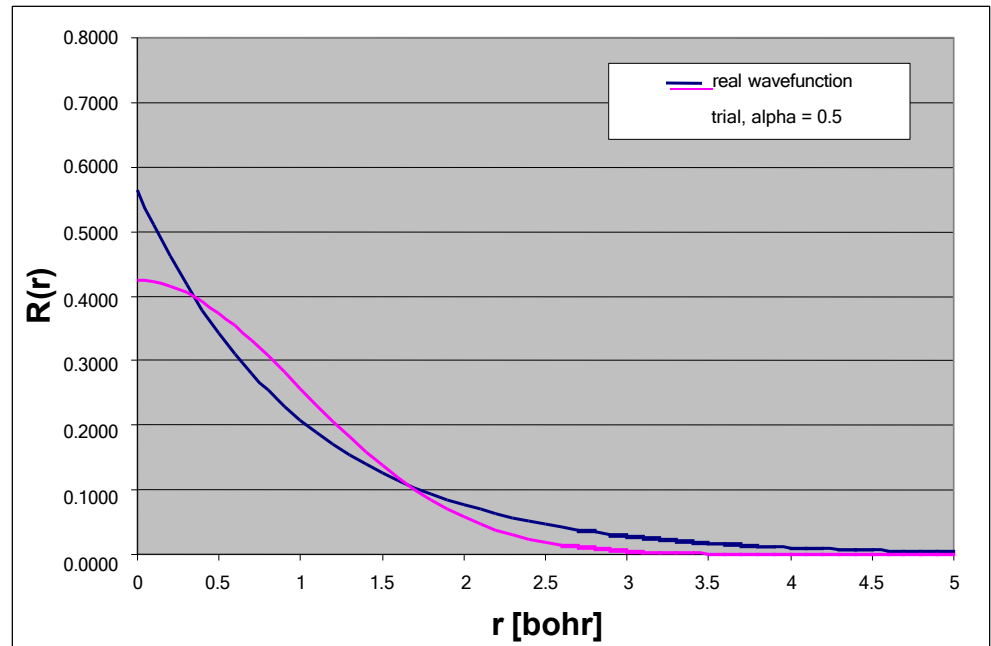
Back to the Hydrogen atom:

a single gaussian function is not a good trial wavefunction!!!

$$\text{real 1s: } \Psi = \left(\frac{1}{\pi} \right)^{1/2} e^{-r}$$

$$\text{trial 1s: } \Phi = \sum_{i=1}^1 a_i \left(\frac{2\alpha_i}{\pi} \right)^{3/4} e^{-\alpha_i r^2}$$

$$a_1=1.0, \alpha_1=0.5$$



Construction and Optimization of Trial Wavefunctions

trial function is a linear combination of basis functions

$$\Phi = \sum_{i=1}^N a_i \phi_i$$

Back to the Hydrogen atom:

a linear combination of gaussian functions is a good trial wavefunction!!!

$$\text{real 1s: } \Psi = \left(\frac{1}{\pi} \right)^{1/2} e^{-r}$$

$$\text{trial 1s: } \Phi = \sum_{i=1}^3 a_i \left(\frac{2\alpha_i}{\pi} \right)^{3/4} e^{-\alpha_i r^2}$$

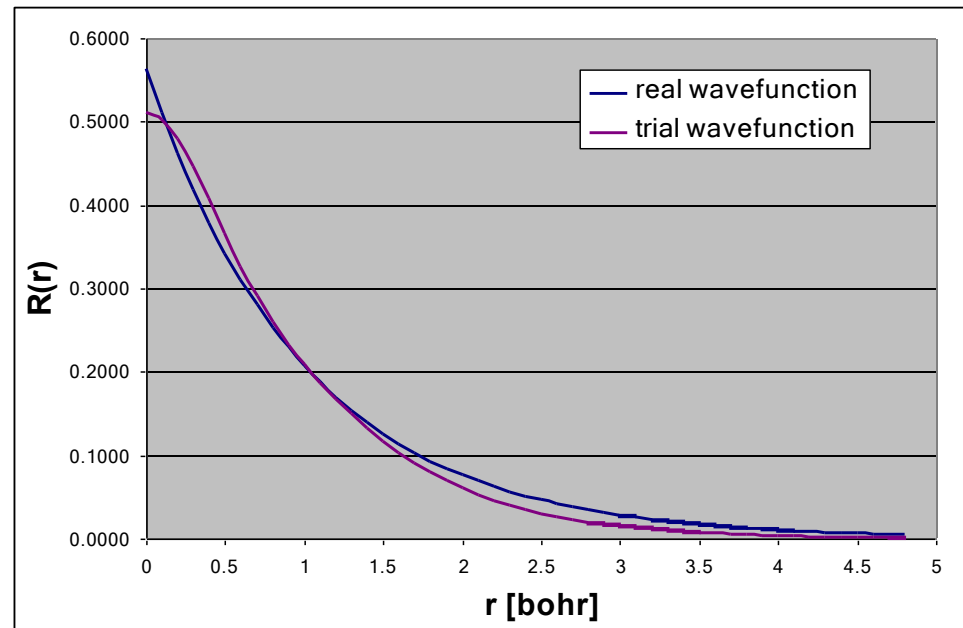
$$a_1=0.1, \alpha_1=3.5$$

$$a_2=0.5, \alpha_2=0.6$$

$$a_3=0.4, \alpha_3=0.2$$

optimized to
minimize energy

fixed to define the basis function



Linear Variation

the best trial wavefunction minimizes the energy

to „optimize“ Φ , we need a relationship between E and $\{a_i\}$

$$E = \frac{\int \Phi^* \hat{H} \Phi d\tau}{\int \Phi^* \Phi d\tau} = \frac{\int \left(\sum_{i=1}^N a_i^* \phi_i^* \right) \hat{H} \left(\sum_{j=1}^N a_j \phi_j \right) d\tau}{\int \left(\sum_{i=1}^N a_i^* \phi_i^* \right) \left(\sum_{j=1}^N a_j \phi_j \right) d\tau} = \frac{\sum_{i=1}^N \sum_{j=1}^N a_i^* a_j \int \phi_i^* \hat{H} \phi_j d\tau}{\sum_{i=1}^N \sum_{j=1}^N a_i^* a_j \int \phi_i^* \phi_j d\tau}$$

in compact notation:

$$E = \frac{\sum_{i=1}^N \sum_{j=1}^N a_i^* a_j H_{ij}}{\sum_{i=1}^N \sum_{j=1}^N a_i^* a_j S_{ij}}$$

$$H_{ij} = \int \phi_i^* \hat{H} \phi_j d\tau$$

$$S_{ij} = \int \phi_i^* \phi_j d\tau$$

Linear Variation

$$E = \frac{\sum_{i=1}^N \sum_{j=1}^N a_i^* a_j H_{ij}}{\sum_{i=1}^N \sum_{j=1}^N a_i^* a_j S_{ij}}$$

- energy of the trial wavefunction
- we want to minimize this energy for all possible linear combinations of N basis functions
→ lower energy = better wavefunction
 - H_{ij} and S_{ij} are constants
 - a_i 's are variables

Minimization:

From calculus, a necessary condition for a function (i.e., energy) to be at its minimum is that its derivatives w. r. t. all of its free variables are zero.

$$\frac{\partial E}{\partial a_k} = 0 \quad \text{for all } k$$

Performing this partial differentiation gives rise to N secular equations:

$$\sum_{i=1}^N a_i (H_{ki} - ES_{ki}) = 0 \quad \text{for all } k$$

Secular Equations

$$\sum_{i=1}^N a_i (H_{ki} - ES_{ki}) = 0 \quad \text{for all } k$$

This set of N equations (running over k) involves N unknowns (a_i 's)

From linear algebra, a set of N equations in N unknowns has a non-trivial solution if and only if the determinant formed from the coefficients of the unknowns ($H_{ki} - E_{ki}$) is equal to zero, that is

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} & \cdots & H_{1N} - ES_{1N} \\ H_{21} - ES_{21} & H_{22} - ES_{22} & \cdots & H_{2N} - ES_{2N} \\ \vdots & \vdots & & \vdots \\ H_{N1} - ES_{N1} & H_{N2} - ES_{N2} & \cdots & H_{NN} - ES_{NN} \end{vmatrix} = 0 \quad \text{Secular determinant}$$

- to solve, we need E first
- get E from secular determinant
- the E_j 's can be used to get a set of a_i 's for each state by solving the secular equations

Secular Equations

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} & \cdots & H_{1N} - ES_{1N} \\ H_{21} - ES_{21} & H_{22} - ES_{22} & \cdots & H_{2N} - ES_{2N} \\ \vdots & \vdots & & \vdots \\ H_{N1} - ES_{N1} & H_{N2} - ES_{N2} & \cdots & H_{NN} - ES_{NN} \end{vmatrix} = 0$$

we can get N trial wavefunctions from N basis functions



$$E_0, E_1, E_2, \dots, E_{N-1}$$



$$\left. \begin{aligned} \sum_{i=1}^N a_i (H_{1i} - ES_{1i}) &= 0 \\ \sum_{i=1}^N a_i (H_{2i} - ES_{2i}) &= 0 \\ &\vdots \\ \sum_{i=1}^N a_i (H_{Ni} - ES_{Ni}) &= 0 \end{aligned} \right\}$$



$$\Phi_0 = \sum_{i=1}^N a_{i(0)} \phi_i$$

$$\Phi_1 = \sum_{i=1}^N a_{i(1)} \phi_i$$

$$\vdots$$

$$\Phi_{N-1} = \sum_{i=1}^N a_{i(N-1)} \phi_i$$

Linear Variation - Summary

Procedure:

1. define a trial function by selecting a set of N basis functions : $\Phi = \sum_{i=1}^N a_i \phi_i$
2. For that set of basis functions, evaluate H_{ij} and S_{ij}
3. construct the secular determinant , and determine the N roots E_j of the secular equation.
4. solve $\left\{ \sum_{i=1}^N a_i (H_{1i} - ES_{1i}) = 0 \right\}$ for each of the N values of E_j to determine a_{ij} for that MO.

Results:

set of coefficients $\{a_{ij}\}$ for each state j that give lowest energy for a specific trial wavefunction

Note:

- more basis functions, ϕ_i , gives more coefficients to „tweak“
- leads to a lower energy, and better wavefunction

Variational principle

All the properties of a molecule can be found if the wavefunction is known: *e.g.*, the electron density distribution or the partial atomic charge. The wavefunction can be found by solving the Schrödinger equation.... But the latter is impossible to solve analytically!! We use the variational principle to go around that problem and find approximate wavefunction.

The variation principle is the basic principle to determine wavefunction of complicated molecular systems. The idea is to optimize the coefficient c_A and c_B of the wavefunction, $\Psi = c_A \phi_A + c_B \phi_B$, such that the system is the most stable, *i.e.* the energy is minimal.

Example: The Hydrogen Molecule H-H

- we'll try to find the best coefficients c_i to give to the **trial wavefunction** Ψ in order to approach the true energy.
- assume the wavefunctions for hydrogens ϕ_A and ϕ_B are known.
- we need a wavefunction Ψ for hydrogen molecule. How to build it?

LCAO principle

$$\Psi = c_A \phi_A + c_B \phi_B \longrightarrow H \Psi = E \Psi \longrightarrow E > E_{\text{true}}$$

Aim: Try to find the (c_A, c_B) that minimize the calculated E

Energy in the LCAO approach

$$(1) \quad E = \frac{\int \psi^* H \psi \, d\tau}{\int \psi^* \psi \, d\tau}$$

→ **Numerator:**

$$\begin{aligned} \int \psi^* H \psi \, d\tau &= \int (c_A \phi_A + c_B \phi_B) H (c_A \phi_A + c_B \phi_B) \, d\tau \\ &= c_A^2 \int \phi_A H \phi_A \, d\tau + c_B^2 \int \phi_B H \phi_B \, d\tau + c_A c_B \int \phi_A H \phi_B \, d\tau + c_A c_B \int \phi_B H \phi_A \, d\tau \\ \alpha_A &= \int \phi_A H \phi_A \, d\tau \quad \alpha_B = \int \phi_B H \phi_B \, d\tau \quad \beta = \int \phi_A H \phi_B \, d\tau = \int \phi_B H \phi_A \, d\tau \end{aligned}$$

α_A is a **Coulomb integral**: it is related to the energy of the e⁻ when it occupies A. ($\alpha < 0$)
 β is a **Resonance integral**: it is zero if the orbital don't overlap. (at R_e , $\beta < 0$)

$$\rightarrow \int \psi^* H \psi \, d\tau = c_A^2 \alpha_A + c_B^2 \alpha_B + 2c_A c_B \beta \quad (1)$$

→ **Denominator:**

$$\begin{aligned} \int \psi^2 \, d\tau &= \int (c_A \phi_A + c_B \phi_B)^2 \, d\tau \\ &= c_A^2 \int \phi_A^2 \, d\tau + c_B^2 \int \phi_B^2 \, d\tau + 2c_A c_B \int \phi_A \phi_B \, d\tau \\ &= c_A^2 S_{AA} + c_B^2 S_{BB} + 2c_A c_B S_{AB} \end{aligned}$$

$$E = \frac{c_A^2 \alpha_A + c_B^2 \alpha_B + 2c_A c_B \beta}{c_A^2 S_{AA} + c_B^2 S_{BB} + 2c_A c_B S_{AB}} \quad (1)$$

How to get the secular equations?

$$E = \frac{c_A^2 \alpha_A + c_B^2 \alpha_B + 2c_A c_B \beta}{c_A^2 S_{AA} + c_B^2 S_{BB} + 2c_A c_B S_{AB}}$$

Let's find the coefficients c_A and c_B to stabilize the energy of the system w.r.t. both c_A and c_B

Let's start with $\partial E / \partial c_A = 0$

$$c_A^2 \alpha_A + c_B^2 \alpha_B + 2c_A c_B \beta = E(c_A^2 S_{AA} + c_B^2 S_{BB} + 2c_A c_B S_{AB})$$

$$c_A^2(\alpha_A - ES_{AA}) + c_B^2(\alpha_B - ES_{BB}) + 2c_A c_B(\beta - ES_{AB}) = 0$$

$$2c_A \alpha_A - 2c_A ES_{AA} + 2c_B \beta - 2c_B ES_{AB} = 0$$

$$c_A(\alpha_A - ES_{AA}) + c_B(\beta - ES_{AB}) = 0$$

$$S_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases} \longrightarrow \text{Hückel approximation}$$

$$\therefore c_A(\alpha_A - E) + c_B \beta = 0$$

Let's continue with $\partial E / \partial c_B = 0$

we arrived at a system with two simultaneous equations:

$$c_A(\alpha_A - E) + c_B\beta = 0$$

$$c_A\beta + c_B(\alpha_B - E) = 0$$



Secular equations

In order to have a solution, other than the trivial solution $c_A = c_B = 0$, we must have:

$$\begin{vmatrix} \alpha_A - E & \beta \\ \beta & \alpha_B - E \end{vmatrix} = 0$$

Secular determinant should be zero

$$(\alpha_A - E)(\alpha_B - E) - \beta^2 = 0 \quad \longrightarrow \quad \text{The 2 roots give the energies of the molecular orbitals formed from the AOs}$$

$$\begin{vmatrix} \frac{\alpha - E}{\beta} & 1 \\ 1 & \frac{\alpha - E}{\beta} \end{vmatrix} = 0 \quad \xrightarrow{x = \frac{\alpha - E}{\beta}} \quad \begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix} = 0 \quad \Rightarrow \quad x^2 = 1 \quad \Rightarrow \quad x = 1, \quad x = -1$$

In this example the secular equation has two solutions and the H_2 molecule has two MOs optimized by energy:

$$c_1 = 1, \quad c_2 = 1 \quad \Rightarrow \quad \Psi_1 = \phi_A + \phi_B, \quad E = -1$$

$$c_1 = 1, \quad c_2 = -1 \quad \Rightarrow \quad \Psi_2 = \phi_A - \phi_B, \quad E = +1$$

Homonuclear diatomic molecules

1) $\Psi = c_A \phi_A + c_B \phi_B$: with $\phi_A = \phi_B \rightarrow \alpha_A = \alpha_B = \alpha$

$$\longrightarrow (\alpha - E)^2 - (\beta - SE)^2 = 0$$

$$E_+ = \frac{\alpha + \beta}{1 + S} \quad (1)$$

$$E_- = \frac{\alpha - \beta}{1 - S} \quad (2)$$

$$\begin{cases} c_A (\alpha - E) + c_B (\beta - SE) = 0 & (1) \\ c_A (\beta - SE) + c_B (\alpha - E) = 0 & (2) \end{cases}$$

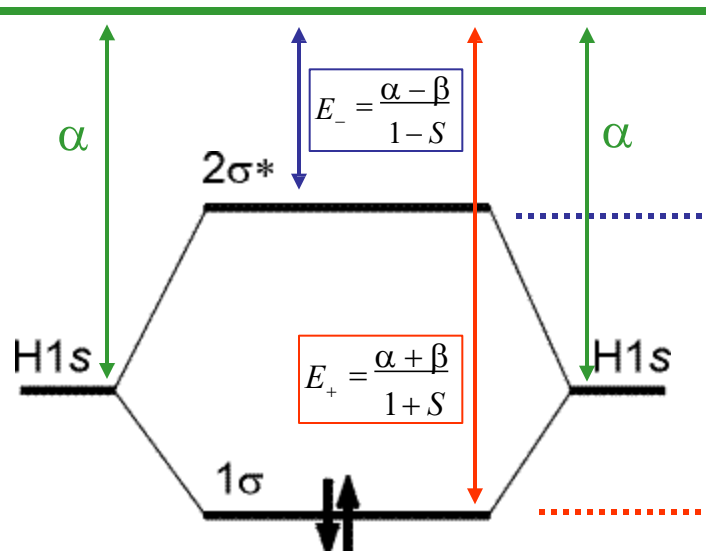
$$c_A = \frac{1}{\{2(1+S)\}^{1/2}} \quad c_B = c_A$$

Ψ_{bonding}

$$c_A = \frac{1}{\{2(1-S)\}^{1/2}} \quad c_B = -c_A$$

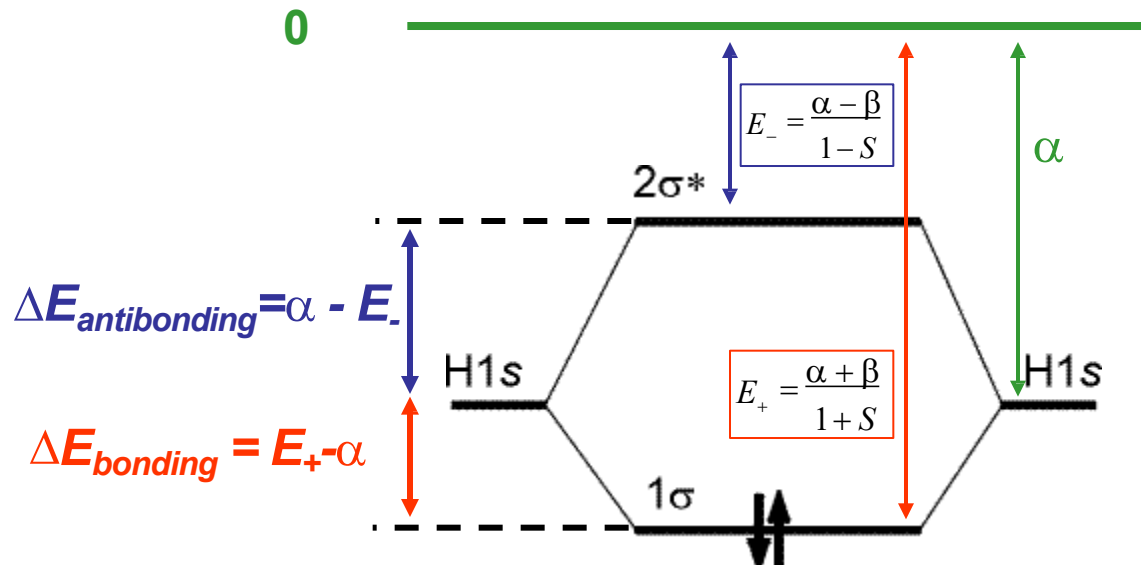
$\Psi_{\text{antibonding}}$

0



$$\Psi_{\text{antibonding}} = \{2(1-S)\}^{-1/2} (\phi_A - \phi_B)$$

$$\Psi_{\text{bonding}} = \{2(1+S)\}^{-1/2} (\phi_A + \phi_B)$$



$$\Delta E_{\text{antibonding}} = \alpha - E_- = \frac{\beta - \alpha S}{1 - S}$$

$$\Delta E_{\text{bonding}} = E_+ - \alpha = \frac{\beta - \alpha S}{1 + S}$$

Since: $0 < S < 1$ * $\Delta E_{\text{antibonding}} >$
 $\Delta E_{\text{bonding}}$

Note 1: He_2 has 4 electrons $\rightarrow$ ground-state configuration: $1\sigma^2 2\sigma^{*2} \rightarrow \text{He}_2$ is not stable!

Note 2: If we neglect the overlap integral ($S=0$), $\Delta E_{\text{antibonding}} = \Delta E_{\text{bonding}} = \beta$
 $\rightarrow$ **The resonance integral β is an indicator of the strength of covalent bonds**

Molecular orbitals of small conjugated molecules

The Hückel approximation

Here, we investigate *conjugated molecules* in which there is an *alternation of single and double bonds* along a chain of carbon atoms.

In the Hückel approach, the π orbitals are treated separately from the σ orbitals, the latter form a rigid framework that determine the general shape of the molecule.

All C are considered similar $\rightarrow$ only *one type of coulomb integral* α for the C2p atomic orbitals involved in the π molecular orbitals spread over the molecule.

The secular determinant

The π *molecular orbitals* are expressed as linear combinations of C2p_z atomic orbitals (LCAO), which are perpendicular to the molecular plane.

♦ *Ethene*, CH₂=CH₂: $\Psi = c_A \phi_A + c_B \phi_B$, where ϕ_A and ϕ_B are the C2p_z orbitals of each carbon atoms.

♦ *Butadiene*, CH₂=CH-CH=CH₂: $\Psi = c_A \phi_A + c_B \phi_B + c_C \phi_C + c_D \phi_D$

The coefficients can be optimized by the same procedure described before: express the total energy E as a function of the c_i and then minimize the E with respect to those coefficients c_i . Inject the energy solutions in the secular equations and extract the coefficients minimizing E .

Following these methods and since $\alpha_A = \alpha_B = \alpha$,
we obtain those secular determinants:

♦ Ethene, $\text{CH}_2=\text{CH}_2$:
$$\begin{vmatrix} (\alpha - E) & (\beta - SE) \\ (\beta - SE) & (\alpha - E) \end{vmatrix} = 0$$

♦ Butadiene:
$$\begin{vmatrix} (\alpha - E) & (\beta_{AB} - S_{AB}E) & (\beta_{AC} - S_{AC}E) & (\beta_{AD} - S_{AD}E) \\ (\beta_{BA} - S_{BA}E) & (\alpha - E) & (\beta_{BC} - S_{BC}E) & (\beta_{BD} - S_{BD}E) \\ (\beta_{CA} - S_{CA}E) & (\beta_{CB} - S_{CB}E) & (\alpha - E) & (\beta_{CD} - S_{CD}E) \\ (\beta_{DA} - S_{DA}E) & (\beta_{DB} - S_{DB}E) & (\beta_{DC} - S_{DC}E) & (\alpha - E) \end{vmatrix} = 0$$

Hückel approximation:

1) All overlap integrals $S_{ij} = 0$ ($i \neq j$).

2) All resonance integrals between non-neighbors, $\beta_{i,i+n} = 0$ with $n > 2$

3) All resonance integrals between neighbors are equal, $\beta_{i,i+1} = \beta_{i+1,i+2} = \beta$

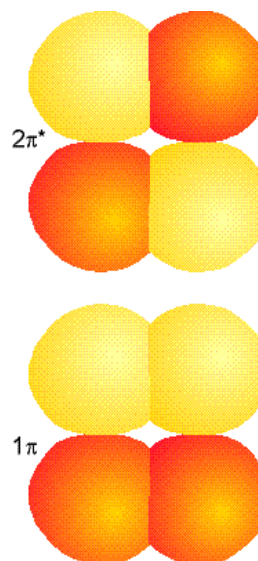
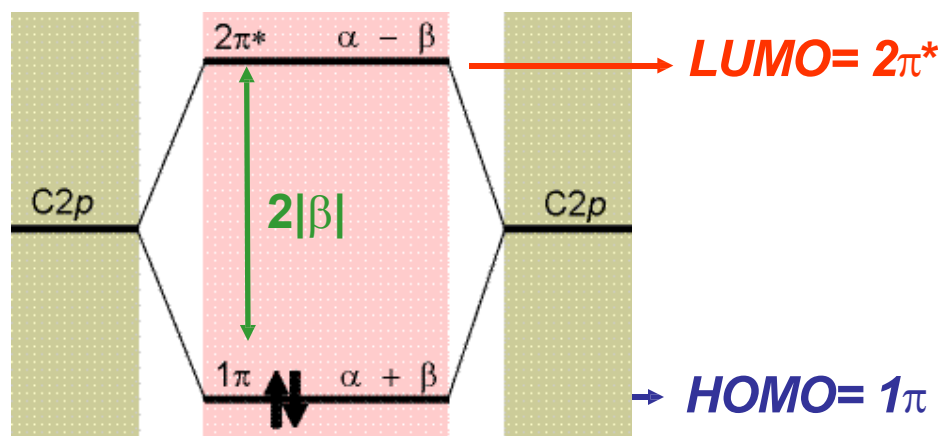
➔ Severe approximation, but it allows us to calculate the general picture of the molecular orbital energy levels.

Ethene and frontier orbitals

Within the Hückel approximation, the secular determinant becomes:

$$\begin{vmatrix} (\alpha - E) & \beta \\ \beta & (\alpha - E) \end{vmatrix} = (\alpha - E)^2 - \beta^2 = 0$$

$E_- = \alpha - \beta$ energy of the **Lowest Unoccupied Molecular Orbital (LUMO)**
 $E_+ = \alpha + \beta$ energy of the **Highest Occupied Molecular Orbital (HOMO)**



→ HOMO and LUMO are the **frontier orbitals** of a molecule.

→ those are important orbitals because they are largely responsible for many **chemical** and **optical properties** of the molecule. Note: The energy needed to excite electronically the molecule, from the ground state $1\pi^2$ to the first excited state $1\pi^1 2\pi^{*1}$ is provided roughly by $2|\beta|$ (β is often around -0.8 eV)

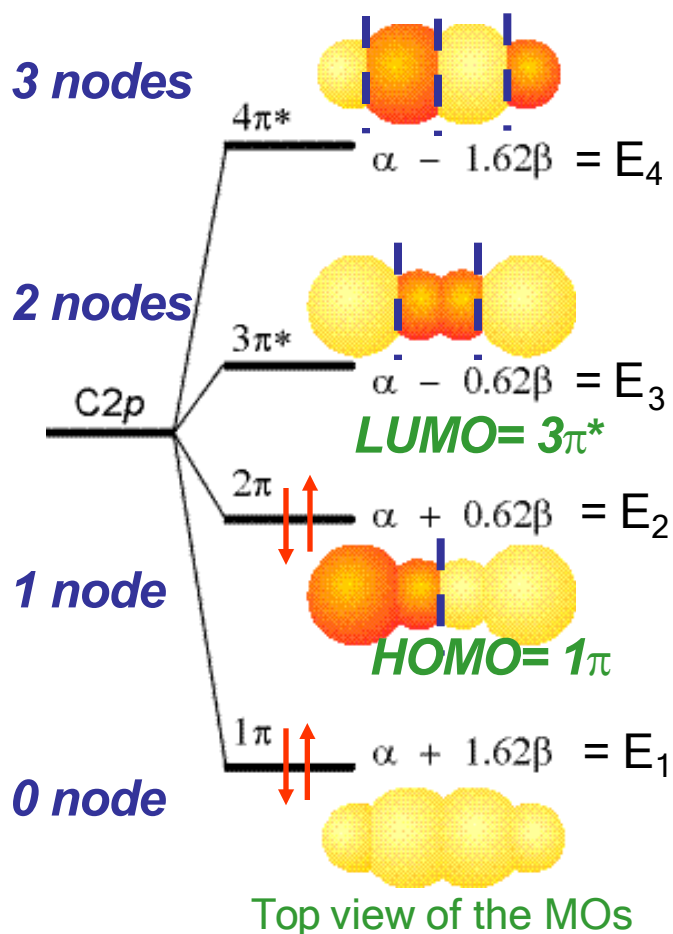
Butadiene and delocalization energy

$$\begin{vmatrix} (\alpha - E) & \beta & 0 & 0 \\ \beta & (\alpha - E) & \beta & 0 \\ 0 & \beta & (\alpha - E) & \beta \\ 0 & 0 & \beta & (\alpha - E) \end{vmatrix} = 0$$

→ 4th order polynomial → 4 roots E

There is 1e⁻ in each 2p_z orbital of the four carbon atoms → 4 electrons to accommodate in the 4 π-type molecular orbitals → the **ground state configuration is 1π² 2π²**

→ **The greater the number of internuclear nodes, the higher the energy of the orbital**



♦ Butadiene C₄H₆: **total π-electron binding energy, E_π is** E_π = 2E₁ + 2E₂ = 4α + 4.48β with two π-bonds

♦ Ethene C₂H₄: E_π = 2α + 2β with one π-bond

♦ Two ethene molecules give: E_π = 4α + 4β for two separated π-bonds.

→ The energy of the butadiene molecule with two π-bonds lies lower by 0.48β (-36kJ/mol) than the sum of two individual π-bonds: this extra-stabilization of a conjugated system is called the **“delocalization energy”**

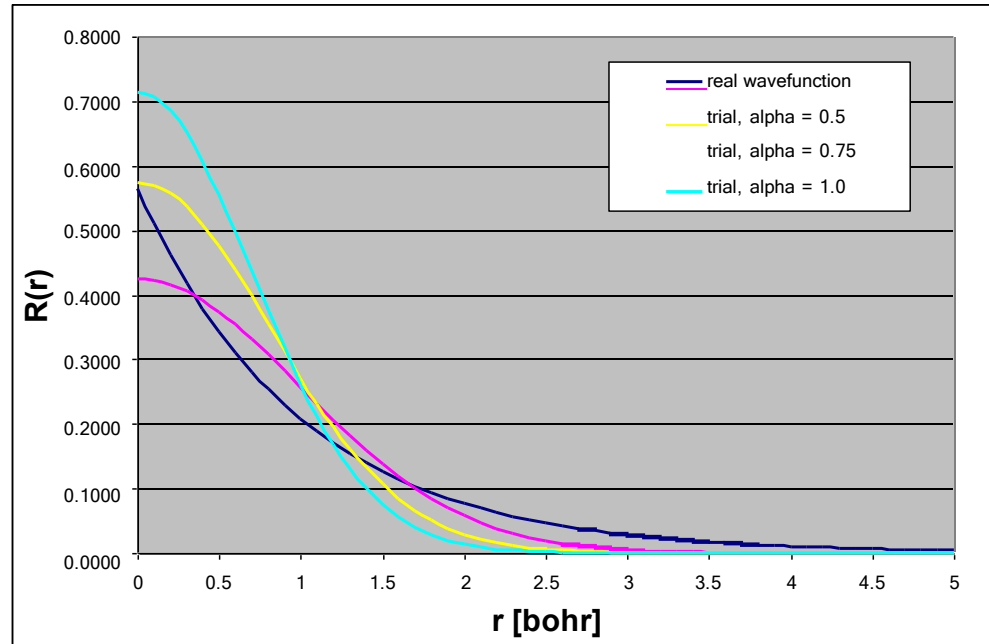
Variational Method

Example for the Hydrogen atom:

$$\text{real 1s: } \Psi = \left(\frac{1}{\pi} \right)^{1/2} e^{-r}$$

$$\text{trial 1s: } \Phi = \left(\frac{2\alpha}{\pi} \right)^{3/4} e^{-\alpha r^2}$$

we want to pick α to give the best agreement with the real solution



Obvious questions:

1. can we really do this? → **variational principle**
2. how do we „optimize“ Φ (i.e. how do we pick α)? → **linear variation method**
3. what form of Φ should we use?

Trial Wavefunctions

how do we select a trial wavefunction for molecules?

1. Assumptions

- each electron occupies a molecular orbital (analogous to atomic orbitals)
- molecular orbitals are orthonormal

2. Constraints

- trial wavefunction must satisfy Pauli exclusion principle
- trial wavefunction must be antisymmetric
- trial wavefunction must not distinguish between electrons

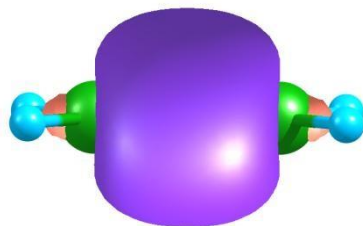
Molecular Orbitals

in atoms:

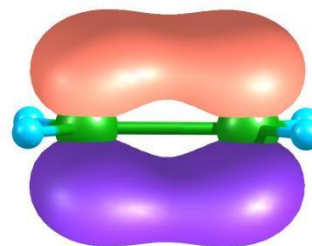
- each electron occupies 1 atomic orbital
e.g. 1s, 2s, 2p_z, ...
- atomic orbitals are centered on nucleus
- atomic orbitals are orthonormal

in molecules:

- each electron occupies 1 molecular orbital
- molecular orbitals can be delocalized across all atoms
- molecular orbitals are orthonormal



C-C σ -bond in ethene



C-C π -bond in ethene

Molecular Orbitals

molecular orbitals depend on spatial coordinates and spin

Spatial orbital, $\psi_i(\mathbf{r})$:

- function of the position vector, $\mathbf{r}$
- describes the spatial distribution of the electron:
$$|\psi_i(\mathbf{r})|^2 d\mathbf{r} = \text{probability of finding electron } i \text{ in } d\mathbf{r}$$
- form an orthonormal set:

$$\int d\mathbf{r} \psi_i^*(\mathbf{r}) \psi_j(\mathbf{r}) = \delta_{ij}$$

Spin functions:

- H_{elec} does not include spin $\rightarrow$ would need a relativistic treatment to get spin
- artificially introduce orthonormal spin functions: $g(\omega) = \alpha(\omega)$ or $\beta(\omega)$

$$\int d\omega \alpha^*(\omega) \alpha(\omega) = \int d\omega \beta^*(\omega) \beta(\omega) = 1$$

$$\int d\omega \alpha^*(\omega) \beta(\omega) = \int d\omega \beta^*(\omega) \alpha(\omega) = 0$$

- ω represent spin coordinates

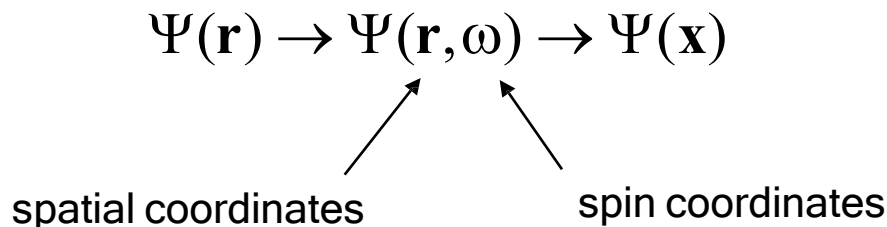
Molecular Orbitals

molecular orbitals depend on spatial coordinates and spin

Spin orbital, $\chi_i(\mathbf{x})$:

$$\chi(\mathbf{x}) = \begin{cases} \psi(\mathbf{r})\alpha(\omega) \\ or \\ \psi(\mathbf{r})\beta(\omega) \end{cases}$$

- spin orbitals are used in quantum chemical calculations
- treatment of spin increases the number of degrees of freedom



- for a system with N electrons, there are $4N$ degrees of freedom

Constraints on the Trial Wavefunction

for variational principle to apply, the trial wavefunction must satisfy the same constraints as the real wavefunction

1. Pauli exclusion principle:

- for atoms → no two electrons can have same set of quantum numbers
- for molecules → no two electrons can be in same state
→ no two electrons can have same $\mathbf{x}_i$

2. Antisymmetry:

- Ψ must be antisymmetric with respect to interchange of coordinates $\mathbf{x}$ between two electrons

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_i, \mathbf{x}_j, \dots, \mathbf{x}_N) = -\Psi(\mathbf{x}_1, \dots, \mathbf{x}_j, \mathbf{x}_i, \dots, \mathbf{x}_N)$$

- if Ψ is antisymmetric, the Pauli exclusion principle will be satisfied

3. Indistinguishability:

- uncertainty principle → we can't know exact positions and velocities of electrons
- Ψ must not distinguish between particles

$$\underbrace{\Psi = 1s(1)2s(2)}_{\text{invalid because electrons 1 and 2 are in specific orbitals}} \quad \text{versus} \quad \Psi = 2^{-1/2} \underbrace{[1s(1)2s(2) - 1s(2)2s(1)]}_{\text{valid because electrons 1 and 2 can be in either orbital}}$$

Molecular Orbitals → Trial Wavefunction

we want the form of the trial wavefunction to closely resemble the true wavefunction

$$\hat{H}_{elec} = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}} + \sum_{i=1}^{N-1} \sum_{j>i}^N \frac{1}{r_{ij}}$$

cannot treat electron-electron terms exactly



let's forget about e-e interactions for now

$$\hat{H}_{eff} = -\sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{I=1}^M \sum_{i=1}^N \frac{Z_I}{r_{Ii}}$$



solve Schrödinger equation

$$\hat{H}_{eff} \tilde{\Psi} = \tilde{E} \tilde{\Psi}$$



solution is product of molecular orbitals

$$\tilde{\Psi}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \mathbf{x}_j, \dots, \mathbf{x}_N) = \chi_1(\mathbf{x}_1) \chi_2(\mathbf{x}_2) \dots \chi_i(\mathbf{x}_i) \chi_j(\mathbf{x}_j) \dots \chi_N(\mathbf{x}_N)$$

Hartree Product Wave Functions

When the only terms in $\hat{H}$ are the one-electron kinetic energy and nuclear attraction terms, the operator is separable:

$$\hat{H} = \sum_{i=1}^N \hat{h}_i$$

Where $\hat{h}_i$ is the one-electron Hamiltonian:

$$\hat{h}_i = -\frac{1}{2} \nabla_i^2 - \sum_{k=1}^M \frac{Z_k}{r_{ik}}$$

Eigenfunctions of $\hat{h}_i$ must satisfy the one-electron Schrödinger equation:

$$\hat{h}_i \chi_i = \varepsilon_i \chi_i$$

Because $\hat{H}$ is separable, its many-electron eigenfunctions can be constructed
As products of one-electron eigenfunctions:

$$\Psi_{HP}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \mathbf{x}_j, \dots, \mathbf{x}_N) = \chi_1(\mathbf{x}_1) \chi_2(\mathbf{x}_2) \dots \chi_i(\mathbf{x}_i) \chi_j(\mathbf{x}_j) \dots \chi_N(\mathbf{x}_N)$$

Slater Determinant Wavefunction

For an N electron system:

$$\Psi^{SD}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \mathbf{x}_j, \dots, \mathbf{x}_N) = (N!)^{-1/2} \begin{vmatrix} \chi_1(\mathbf{x}_1) & \chi_2(\mathbf{x}_1) & \dots & \chi_N(\mathbf{x}_1) \\ \chi_1(\mathbf{x}_2) & \chi_2(\mathbf{x}_2) & \dots & \chi_N(\mathbf{x}_2) \\ \vdots & \vdots & & \vdots \\ \chi_1(\mathbf{x}_N) & \chi_2(\mathbf{x}_N) & \dots & \chi_N(\mathbf{x}_N) \end{vmatrix}$$

- columns labeled by molecular orbitals
- rows labeled by electrons
- shorthand notation: $\Psi^{SD}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \mathbf{x}_j, \dots, \mathbf{x}_N) = |\chi_1 \chi_2 \dots \chi_i \chi_j \dots \chi_N\rangle$

Physics captured:

- electrons of opposite spin are treated independently
→ neglects instantaneous Coulomb interactions
- electrons of same spin are correlated through exchange interactions
→ accounts for Pauli exclusion principle
→ significant improvement over Hartree product

Many body problem:

- ❖ For large interacting system, we first need to consider a many particle wave function.
- ❖ Many body Hamiltonian for electron and nucleus is of the form given below



1926

Schrödinger

Innocent look of wave equation

$$H\psi(r, R, t) = E \psi(r, R, t)$$

M	nuclear	R
m _e	electronic	r
	mixed	

$$H\psi = \underbrace{\left[-\sum_n \frac{\hbar^2}{2M} \nabla_n^2 - \sum_i \frac{\hbar^2}{2m_e} \nabla_i^2 \right]}_{\text{Kinetic terms}} + \underbrace{\left[\frac{1}{4\pi\epsilon_0} \frac{1}{2} \sum_{n \neq m} \frac{Z_n Z_m e^2}{|R_n - R_m|} + \frac{1}{4\pi\epsilon_0} \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|r_i - r_j|} \right]}_{\text{Potential terms}}$$

$\hat{T}_R$
 $\hat{T}_r$

$\hat{V}_r$
 $\hat{V}_{r,R}$

$$H\psi = \hat{T}_r + \hat{T}_R + \hat{V}_r + \hat{V}_{r,R} + \hat{V}_R$$

$$= \underbrace{\hat{T}_R + \hat{V}_R}_{\text{nuclear}} + \underbrace{\hat{T}_r + \hat{V}_r}_{\text{electronic}} + \underbrace{\hat{V}_{r,R}}_{\text{mixed}}$$

Notational convenience:

$$\Psi(r_1, r_2, \dots, R_1, R_2, \dots; t)$$

$$= (\{r_i\}, \{R_i\}; t)$$

$$\Psi(r, R; t)$$

$$\Psi$$

Approximations for solving many body problem

- ❖ The Born-Oppenheimer approximation
- ❖ Hartree approximation
- ❖ Hartree-Fock method
- ❖ Hohenberge- Kohen
- ❖ Kohn-Sham approach (Walter Kohn and Lu.J.Sham)

1. Born-Oppenheimer approximation:

- ❖ The nuclei are much heavier than electrons.
- ❖ They move much more slowly and hence neglect the nuclear kinetic energy.
- ❖ The wave function separated into electronic and nuclear part and determine motion of electrons with nuclei held fixed.

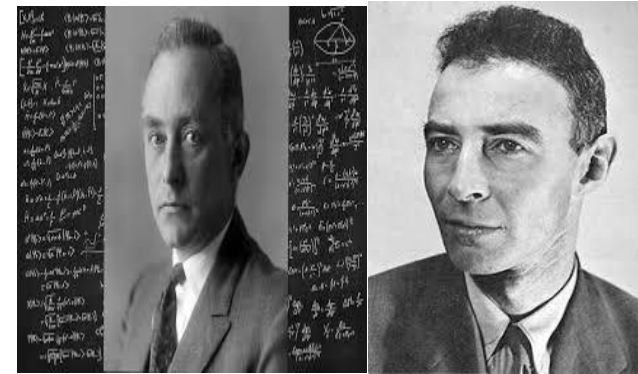
Since the total hamiltonian for electron and nucleus is:

$$\begin{aligned}
 H\Psi &= \hat{T}_r + \hat{T}_R + \hat{V}_r + \hat{V}_{r,R} + \hat{V}_R \\
 &= \underbrace{\hat{T}_R + \hat{V}_R}_{\text{nuclear}} + \underbrace{\hat{T}_r + \hat{V}_r}_{\text{electronic}} + \underbrace{\hat{V}_{r,R}}_{\text{mixed}}
 \end{aligned}$$

then the hamiltonian for the electronic part

$$\begin{aligned}
 H\Psi &\cdot \underbrace{\hat{T}_r + \hat{V}_r}_{\text{electronic}} + \underbrace{\hat{V}_{r,R}}_{\text{mixed}} \\
 &=
 \end{aligned}$$

$$\hat{H}_e = -\sum_i \frac{1}{2} \nabla_i^2 + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|r_i - r_j|} - \sum_{i,n} \frac{Z_n}{|r_i - R_n|}$$

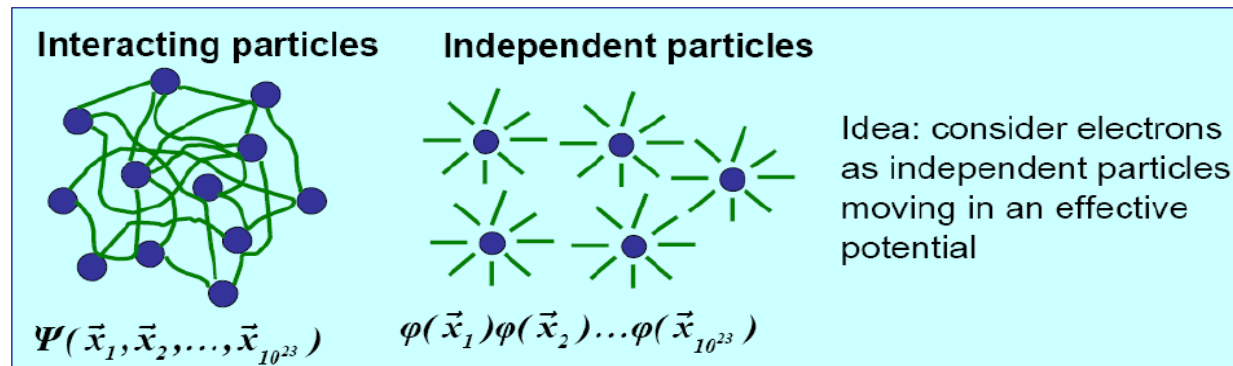


Max Born and Robert Oppenheimer

1927

2. Hartree approximation: One electron model

- ❖ Reduce the complexity of electron-electron interactions.
- ❖ Electrons are independent and interacts with others in an averaged way.
- ❖ For an n-electron system, each electron does not recognize other as single entities but as a mean field.
- ❖ Hence, n-electron system becomes a set of non-interacting one-electron system where each electron moves in the average density of rest electrons.



1928

Self-consistent field procedure to solve the wave equation:

$$\hat{H}_e = -\sum_i \frac{1}{2} \nabla_i^2 + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|r_i - r_j|} - \sum_{i,n} \frac{Z_n}{|r_i - R_n|}$$

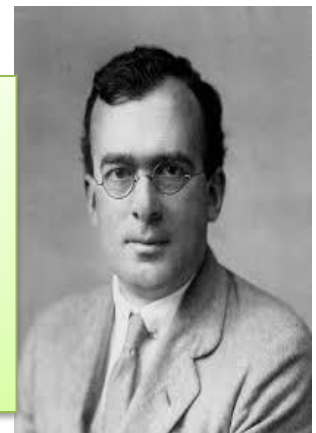
$$\left(-\frac{1}{2} \nabla_i^2 + V_H + V_{\text{ext}} \right) \Psi(\mathbf{r}) = E \Psi(\mathbf{r})$$

$$E = E_1 + E_2 + E_3 + \dots + E_n$$

R-nuclear
r- electron

V_{ext} = electron and nuclei interaction potential

V_H = Hartree potential (e-e interaction)



Hartree

Hartree method produced crude estimation of energy due to two oversimplifications:

- ✓ *Hartree method does not follow two basic principles of quantum mechanics: the antisymmetry principle and Pauli's exclusion principle.*
- ✓ *Does not count the exchange and correlation energies coming from n-electron nature.*



Hartree

The Hartree method, therefore, was soon refined into the Hartree-Fock method.....continue...

3. Hartree-Fock method

Based on the one-electron and mean-field approach by Hartree, V.A. Fock enhanced the methods to higher perfection. Fock and J.C. Slater in 1930 generalised the Hartree's theory to take into account the antisymmetry requirement.

- ❖ In HF method, the n-electron wave function approximated as a linear combination of non-interacting one-electron wave function in the form of Slater determinant.

$$\Psi(1,\dots,N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(1) & \psi_1(2) & \cdots & \psi_1(N) \\ \psi_2(1) & \psi_2(2) & \cdots & \psi_2(N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_N(1) & \psi_N(2) & \cdots & \psi_N(N) \end{vmatrix}$$

❖ Slater
determinant

1930



Fock

$$\left(-\frac{1}{2}\nabla^2 + v_{\text{ext}}(r) + v_H(r) \right) \psi(r) = E \psi(r)$$

$$E = E_{\text{kin}} + E_H + E_{\text{ext}} + E_x$$

$V_H = V_{ij}$ Hartree or Coulomb interaction energy of two electrons

E_x = Exchange energy comes from the antisymmetric nature of wave function in the Slater determinant.

Difficulties with Hartree-Fock Theory:

- Correlation energy and
- Problem of dealing 3N dimensional .

A new approach has been developed known as **Density Functional Theory (DFT)**.

- In 1964 Hohenberg and Kohn showed that schrodinger equation (3N dimensional e.g. 10 electrons require 30 dimensions) could be reformulated in terms of **electron density $n(r)$** with non-interacting n separate 3-dimensional ones.
- The main objective of DFT is to replace **the many-particle electronic wavefunction** with **the electron density** as the basic quantity.
- The electron density $n(r)$, the central player in DFT decides everything in an n -electron quantum state where there is **no individual electron density** but a **3-dimensional density of electrons**.
- The addition of *all the electron densities over the whole space* naturally return to *the total number of electrons* in the system.
- The knowledge of overlapping of atomic electron density, roughly generate the electron density of the solids.
- This theory gives approximate solutions to both **Exchange** and **Correlation** Energies.

The Fundamental Pillars of DFT

First Hohenberg Kohn (HK) theorem: *The ground-state energy is a unique functional of the electron density $n(r)$.*

- This theorem provides one to one mapping between ground state wave function and ground state charge density.
- The ground state charge density can uniquely describe all the ground state properties of system.
- The fundamental concept behind density functional theory is that charge density (3-Dimensional) can correctly describe the ground state of N-particle instead of using a wave function (3N-Dimensional).

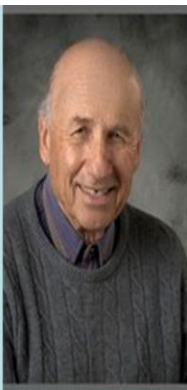
Second Hohenberg Kohn (HK) theorem: *The electron density that **minimizes the energy** of the overall functional is the **true electron density**.*

- If the true functional form of energy in terms of density gets known, then one could vary the electron density until the energy from the functional is minimized, giving us required **ground state density**.
- This is essentially a **variational principle** and is used in practice with approximate forms of the functional.
- The simplest possible choice of a functional can be a constant electron density all over the space.



Pierre C
Hohenberg
Kohn 1923,
german-born
american

DFT



Walter Kohn
1923,
Austrian-born
American
nobel 1998

Two Hohenberg and
Kohn theorems :

The existence of a unique functional.

The variational principle.

5. Kohn- Sham Approach (1965):

➤ KS replace the interacting n-electron system with a system of one-electron (non-interacting) system in effective potential having the **same ground state**.

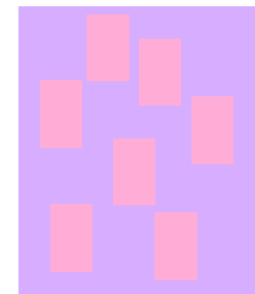
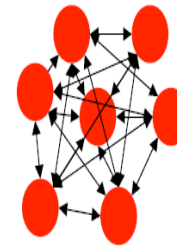
since the kinetic energy; $\mathbf{E} = \mathbf{E}_{\text{kin}}^{\text{int}} + \mathbf{E}_{\text{ext}} + \mathbf{E}_{\text{H}} + \mathbf{E}_{\text{x}} + \mathbf{E}_{\text{c}}$

where $\mathbf{E}_{\text{kin}} = \mathbf{E}_{\text{kin}}^{\text{non}} + \mathbf{E}_{\text{kin}}^{\text{int}}$

→ Replace the original many-body problem with an auxiliary system of non-interacting particles.

$$\mathbf{E} = \mathbf{E}_{\text{kin}}^{\text{non}} + \mathbf{E}_{\text{kin}}^{\text{int}} + \mathbf{E}_{\text{ext}} + \mathbf{E}_{\text{H}}^{\text{int}} + \mathbf{E}_{\text{x}} + \mathbf{E}_{\text{c}}$$

Interacting electrons
+ real potential



Non-interacting
auxiliary
particles in an
effective potential

$$\mathbf{E} = \mathbf{E}_{\text{kin}}^{\text{non}} + \mathbf{E}_{\text{ext}} + \mathbf{E}_{\text{H}} + \mathbf{E}_{\text{xc}} = \mathbf{F} [\mathbf{n}(\mathbf{r})] + \mathbf{E}_{\text{ext}}$$

Walter Kohn

Lu Jeu Sham



The Kohn-Sham equations have the form

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}}(\mathbf{r}) \right) \psi_{\mu}(\mathbf{r}) = \varepsilon_{\mu} \psi_{\mu}(\mathbf{r})$$

where

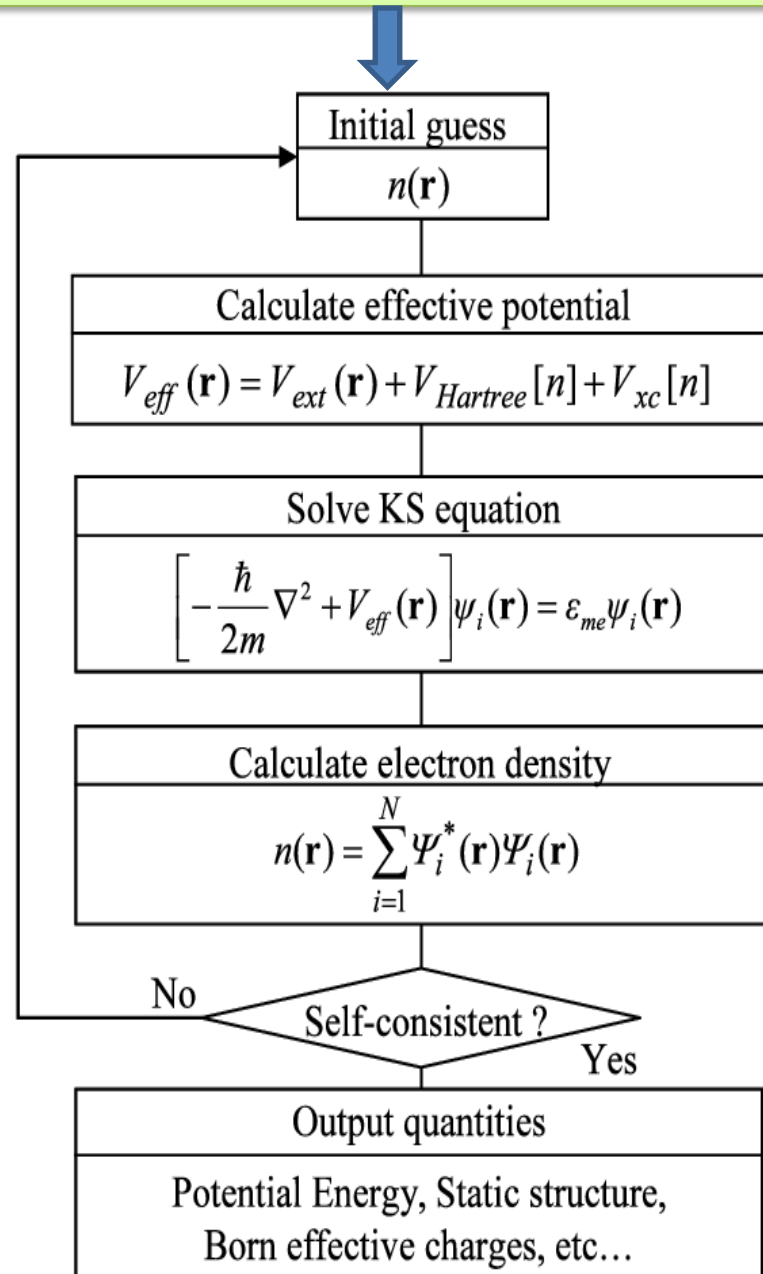
$$V_{\text{eff}}(\mathbf{r}) = V(\mathbf{r}) + V_{\text{H}}(\mathbf{r}) + V_{\text{XC}}(\mathbf{r})$$

DFT in Practice: Kohn-Sham Self Consistency loop

Hence final KH equation has the form:

$$\hat{H}_{KS}\psi_i(\mathbf{r}) = \left[-\frac{1}{2}\nabla^2 + v_H(\mathbf{r}) + v_{xc}(\mathbf{r}) + v_{ext}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r})$$

$$v_H(\mathbf{r}) = \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad v_{xc}(\mathbf{r}) = \frac{\delta E_{xc}}{\delta n(\mathbf{r})} \quad n(\mathbf{r}) = \sum_{i=1}^N |\psi_i(\mathbf{r})|^2$$



Exchange-correlation approximation

1. Local density approximation (LDA)

- Approximation used to find out exchange-correlation function.
- Exchange-correlation energy functional is purely local.
- Ignores corrections to the exchange-correlation energy at a point $\mathbf{r}$ due to nearby inhomogeneities in the electron density.

2. Generalized Gradient Approximation (GGA)

- Depends on local density and its gradient.
- GGA uses information about the local electron density and also the local gradient in the electron density.

Though GGA includes more physical information than LDA

It is not necessary that it must be more accurate.

There are large number of distinct GGA functionals depending on the ways in which information from the gradient of the electron density can be included in a GGA functional.

DFT in Practice: The exchange-correlation Functional

$$E_{xc}[n] = \int d\mathbf{r} n(\mathbf{r}) \varepsilon_{xc}[n(\mathbf{r})]$$

$$v_{xc} = \frac{\delta \epsilon_{xc}}{\delta n}$$

potential

Local density approximation (LDA)

$$E_{xc}^{\text{LDA}}[n] = \int d\mathbf{r} n(\mathbf{r}) \varepsilon_{xc}^{\text{HEG}}(n(\mathbf{r}))$$

where $\varepsilon_{xc}^{\text{HEG}}(n)$ is the exchange-correlation energy of a homogeneous electron gas of density n

Generalized gradient approximations (GGA; semi-local functionals)

$$E_{xc}^{\text{GGA}}[n] = \int d\mathbf{r} n(\mathbf{r}) \varepsilon_{xc}(n(\mathbf{r}), \Delta n(\mathbf{r}))$$